

WEAREDEVELOPERS X BAND: DARK FACTORY HACKATHON

Pocketful

Autonomous Zero-Sum Fintech Engine & Bitemporal Ledger

A clean-room enterprise financial ledger synthesized 100% autonomously inside the BAND Desktop multi-agent factory without human steering. Built with formal zero-sum conservation, SHA-256 audit chaining, and offline containerization.

 <https://pocketful-fintech.vercel.app>

 <https://github.com/fokrulanthro16-eng/pocketful-submission>

 <https://youtu.be/70vY0ZwfjpM>

What is BAND Desktop?

The Autonomous Multi-Agent Desktop Runtime & Software Factory by Thenvoi.

CORE HACKATHON PLATFORM

⚡ Platform Definition & Architecture

BAND Desktop is an advanced local-first desktop environment engineered to coordinate autonomous, decoupled LLM agent workflows into a unified software delivery factory.

Key primitives of BAND Desktop:

- **Autonomous Rooms:** Isolated collaboration chambers where specialized agents interact in a shared conversational runtime.
- **Decoupled Seats:** Modular agent nodes (Planner, Developer, Verifier) powered by high-reasoning frontier models (e.g., DeepSeek-R1).
- **Standing Generic Mandates:** Pure behavioral and structural contracts governing agent handoffs with 0% track bias.

🏭 The Dark Factory Paradigm

A **Dark Factory** operates without human intervention or steering. Humans provide only the initial specification kickoff; the multi-seat BAND cluster autonomously executes the entire development lifecycle:

- **Autonomous Task Breakdown:** Isolating API contracts & data invariants.
- **Clean-Room Code Synthesis:** Writing backend services and Docker assets.
- **Automated Gating & Certification:** Running stress test harnesses to certify production readiness before gate handoff.

Mathematical Invariant & Financial Core

FORMALLY VERIFIED

Strict zero-sum conservation eliminates balance drift and race conditions.



Absolute Zero-Sum Invariant

Traditional ledgers rely on eventual consistency, causing critical balance leakage during concurrent transactions.

Pocketful enforces a mathematically bounded invariant:

$$\sum \Delta \text{Balances} + \sum \Delta \text{Holds} + \sum \Delta \text{Escrow} = 0$$

For every credit event, an exact matching debit is atomically settled across user ledgers and system reserves. Overdrafts return strict HTTP 400 rejections.



Idempotency & Mutex Locking

- **Reentrant Concurrency Mutex:** Guarantees serialized read-then-write cycles across concurrent mutation requests.
- **Mandatory Idempotency Keys:** Repeated transaction dispatches produce cached, deterministic outputs without duplicate balance deductions.
- **Bitemporal Auditing:** Tracks both transaction valid-time and system ledger assertion-time.

3-Seat Execution & Autonomous Self-Healing

50% SCORING RUBRIC

Live detection, release halting, and automated remediation inside BAND Desktop.

Decoupled 3-Seat Topology

- 1. Systems Planner:** Isolates mathematical invariants, data schemas, and API contracts.
- 2. Core Developer:** Implements FastAPI ledger, SQLite WAL journaling, and container.
- 3. Quality Verifier:** Executes 100-worker concurrency stress harnesses.

Intercepted Race Condition & Self-Healing Loop

During Stage-1 synthesis, the Verifier detected a race condition on request #27 causing balance to dip to -\$25.00.

 Release Gate Halted: Defect trace returned autonomously to Developer.

 Self-Healing Loop: Developer injected asyncio mutex locks and WAL serialized mode. 100% passed upon re-audit.

Forensic Audit Chain & Real-Time AML Sentinel

ENTERPRISE SECURITY

Cryptographic non-repudiation and pre-settlement fraud heuristics.

SHA-256 Cryptographic Hash Chaining

Every state transition is immutably anchored to the previous block:

```
Hash_n = SHA256(Hash_(n-1) || Tx_Payload || UTC_Timestamp)
```

Provides institutional auditors with tamper-proof mathematical certainty against retroactive ledger modifications or historical manipulation.

Real-Time AML Sentinel Monitor

Operates concurrently with the ledger pipeline:

- **Velocity Anomaly Detection:** Flags rapid burst transfers exceeding velocity parameters.
- **Micro-Structuring Flags:** Identifies repeated transfers structured just below regulatory limits (\$10,000 threshold).
- **Pre-Settlement Interception:** Freezes anomalous payloads before balance commit.

Verification Matrix & Final Deliverables

GATE 1 - 4 CERTIFIED

Complete compliance across all hackathon gates and disqualification checks.

 **193 / 193 Tests Passed (Contiguous Regression-Free)**

38/38

STAGE 1: LEDGER CORE

47/47

STAGE 2: HOLD ESCROW

52/52

STAGE 3: SPLIT PAYMENTS

56/56

STAGE 4: AML & AUDIT

Zero-Network Offline Container

Standalone multi-stage Docker container (Python 3.11-slim) validated on port 8080 with 0 outbound network egress.

Verified Deliverables

Live App: pocketful-fintech.vercel.app

Video Demo: <https://youtu.be/70vY0ZwfjpM>